

C++ Programming Course Content

1. Introduction to C++

- History and features of C++
- Installing compiler and IDE (GCC, Visual Studio, CodeBlocks)
- Structure of a C++ program
- Compilation and execution process

2. Basic Concepts

- Variables and data types
- Constants and literals
- Operators (Arithmetic, Relational, Logical, Bitwise, Assignment)
- Input/Output (cin, cout)

3. Control Statements

- if, if-else, nested if
- switch-case
- Loops: for, while, do-while
- break, continue, goto

4. Functions

- Function declaration and definition
- Call by value and call by reference
- Inline functions
- Recursion

5. Arrays and Strings

- One-dimensional and multi-dimensional arrays
- String handling using char arrays
- String class in C++

6. Pointers

- Pointer basics
- Pointer arithmetic
- Pointers and arrays

- Dynamic memory allocation (new, delete)

7. Object-Oriented Programming (OOP)

- Classes and objects
- Constructors and destructors
- Encapsulation and data hiding
- Access specifiers (public, private, protected)

8. Inheritance

- Single, multiple, multilevel, hierarchical inheritance
- Method overriding
- Virtual functions and polymorphism

9. Polymorphism

- Function overloading
- Operator overloading
- Runtime polymorphism

10. Templates and STL

- Function templates
- Class templates
- Standard Template Library (STL): vector, list, map, set, stack, queue

11. Exception Handling

- try, catch, throw
- Multiple catch blocks
- User-defined exceptions

12. File Handling

- Reading and writing files
- fstream, ifstream, ofstream

13. Advanced Topics

- Namespaces
- Lambda expressions
- Smart pointers

- Multithreading basics

14. Mini Projects

- Student Management System
- Bank Management System
- Library Management System